MUHAMMAD BIN AAMIR

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EDUCATION

Caldwell University — Caldwell, NJ *Expected Graduation: Dec 2026*

Bachelor of Science in Computer Science | ***GPA: 3.8***

- **Honors & Awards:** CU Presidential Scholarship, St. Thomas Aquinas Award, Caldwell Honors Program, Sienna CU Grant
- **Relevant Coursework:** Data Structures & Algorithms, Database Design, Network & Operating Systems, Applied Machine Learning / AI Fundamentals, Software Engineering

TECHNICAL SKILLS

Languages & Frameworks:	Python, SQL, JavaScript, FastAPI, React, SQLAlchemy, HTML/CSS
ML & Data Tools:	NumPy, Pandas, Scikit-learn (Coursework/Projects), Dataset Preprocessing, Data Modeling
Databases & Tools:	PostgreSQL, Alembic, REST APIs, Git/GitHub, Linux Environments, VS Code
Core Competencies:	System Architecture, Database Design, Concurrency & Security Logic, Data Pipelines, Algorithmic Optimization

PROJECTS & TECHNICAL WORK

Facilities Ticketing & Operations Platform *Senior Capstone Project*

- Designed and engineered a full-stack digital facilities management platform to handle operational request intake, ticket routing workflows, and role-based access control.
- Engineered relational schemas in PostgreSQL/SQL to model request lifecycles, operational statuses, and equipment asset tracking.
- Implemented secure authentication and authorization controls to restrict administrative permissions and maintain data integrity across endpoints.
- Applied software engineering methodologies to validate input security, address boundary conditions, and optimize operational system performance.

Multi-Stop Route Optimization (MSRO) *NASA Space Hackathon Honorable Mention*

- Developed an algorithmic pathfinding solution solving a Traveling Salesperson Problem (TSP) variant using Google Maps API integration.
- Implemented data optimization logic to minimize transit costs, algorithmic overhead, and time constraints across multi-stop routing parameters.

Car Fuel Log System & Gym Database Systems *Personal / Coursework Projects*

- Built full-stack relational database systems (SQL/Python/JS) featuring custom CRUD logic, transaction management, and robust input validation to ensure dataset consistency.

EXPERIENCE

Software Engineering Intern *Jan 2025 – Present*

Ummah Professionals (intern-net)

- Identified and resolved a critical database race condition in slot-booking logic, preventing concurrency vulnerabilities and data corruption under high concurrent request volume.
- Designed and deployed structured REST APIs for user intake, availability management, and automated third-party integrations (Google Meet API).
- Consolidated complex transactional booking flows into unified API endpoints, implementing 24-hour verification logic to reduce scheduling conflicts.
- Led pull request (PR) reviews and API repository maintenance, deprecating insecure or redundant endpoints to ensure system reliability.

Technology & Operations Associate *Dec 2023 – Present*

Campus Safety, Caldwell University

- Designed, built, and deployed a digital Visitor Registration System prototype to digitize paper logs and strengthen physical campus security.
- Streamlined operational data intake procedures, reducing administrative overhead by 15% while improving record accuracy and accessibility.

Facilities Operations Intern (SSC) *Summer 2024 & Winter 2024*

Caldwell University

- Supported digital record-keeping for campus maintenance infrastructure, managing asset records and operational data to inform facility improvements.

LEADERSHIP & ACTIVITIES

Re-founder & President *Jan 2024 – Sep 2024*

Muslim Student Association (MSA), Caldwell University

- Re-established the campus organization, expanding membership to 45+ active students in the first semester while directing operations, budget allocation, and event planning.